

WHO and diabetes

by Hilary King

The highest prevalence rates of diabetes are to be found in developing nations. In extreme cases one-third or more of the adult population suffers from this disease, and one-tenth is not unusual in urban areas of the developing world. Diabetes can no longer be considered a disease of affluent nations alone; it has become a global problem, a major epidemic of the late twentieth century, and one which shows no sign of abating.

Chronic noncommunicable diseases arise as major killers and causes of disability in many areas of the world. WHO is developing several types of approaches to tackle them. Firstly, there is the study of the diseases themselves—cancer, cardiovascular diseases, rheumatic diseases, diabetes mellitus—and of the ways that exist to control them. Secondly, programmes concentrate on different risk factors which confront people in their social and cultural environment, and which may be detrimental to their health, such as tobacco or alcohol. Thirdly, different programmes seek to promote health in particular groups of society, such as schoolchildren, the elderly and the workforce. Finally, WHO in 1990 launched the Inter-Health programme, which is an integrated approach to explore the possibility of controlling

simultaneously several risk factors like obesity, lack of exercise and inappropriate diet.

Diabetes has a strong genetic component, but it also stems from social, economic, behavioural and environmental factors. There may be fundamental differences between developed and developing countries in this respect, but there is also a predictable pattern of health change in most populations where infectious diseases tend to decline as economic conditions improve and noncommunicable diseases become major causes of disease and death. In some cases, high rates of infections and noncommunicable diseases coexist, leading to the so-called "double burden".

Diabetes, particularly the noninsulin-dependent form, is one of the most important noncommunicable diseases to have emerged with industrialization and the subsequent changes brought about in modern societies.

The three principal components of WHO's strategy to tackle the rising rates of diabetes are (1) research, to discover the distribution and determinants of the disease, and new technology for its treatment; (2) education, to prevent the disease from developing unnecessarily and to help diabetic children or adults to live with their disease; and (3) strengthening national health services, to control the disease

more efficiently. WHO will act as a catalyst, by providing the expertise and resources needed to initiate national programmes. Strong commitment on the part of the national government concerned is also required.

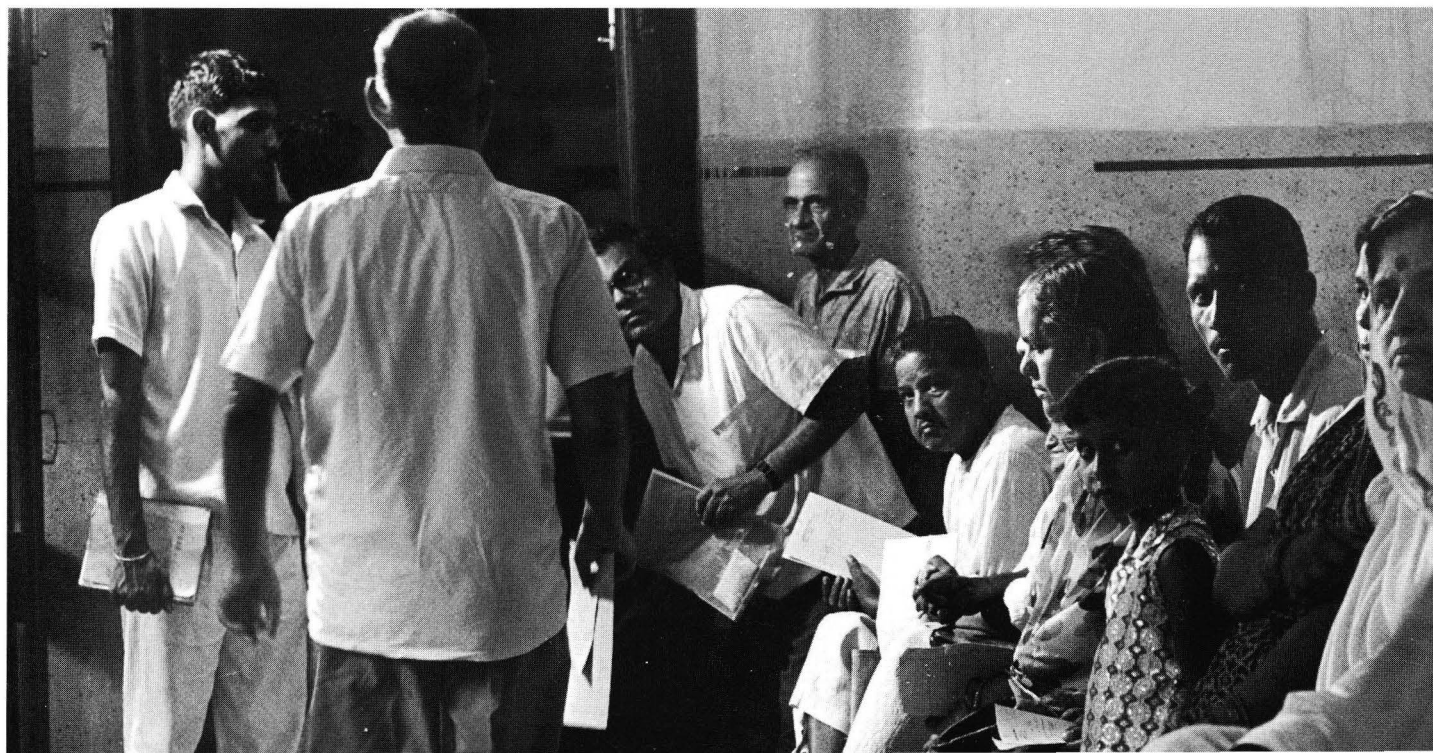
A World Health Assembly resolution in May 1989 invited Member States of WHO to assess the national importance of the disease and to take measures to control it. To assist in this process, WHO recently coordinated the formulation of guidelines for the development of national diabetes programmes. These will be made available to health planners in all countries and will also be used in training courses on management of services and programmes for diabetes.

A vigorous campaign to strengthen national programmes in European countries is being put into effect by WHO's Regional Office for Europe, in collaboration with the International Diabetes Federation (IDF). The St Vincent Declaration on diabetes research and care in Europe has set goals and five-year targets. It is hoped that similar initiatives may be possible in other regions in the future. The first World Diabetes Day, co-sponsored by WHO and the IDF, will be held on 27 June 1991. It will draw unprecedented attention to persons with diabetes all over the world.

Research is vital

The frequency of diabetes and its severe complications need to be fully documented. Epidemiological informa-

Waiting-room of a diabetes clinic in India. A big stumbling-block to controlling diabetes in the Third World is case-finding.





tion is especially valuable at a time when there is evidence that the prevalence and incidence of the disease are on the increase in many parts of the world. But epidemiology serves other purposes: to seek information on the causes of the disease with a view to preventing it; to determine the factors that may retard, or on the contrary accelerate, the development of complications; to evaluate the effectiveness of treatments and public health interventions; and to provide information for the rational planning of health services.

Back in 1970, the WHO Multinational Study of Vascular Disease in Diabetics investigated apparent differences in rates of vascular complications of diabetes. More than 6500 diabetics were recruited in 13 countries around the world to participate in this project, which was coordinated by a London-based WHO collaborating centre for the study of long-term complications of diabetes. While the study on complications of diabetes is now over, the follow-up on mortality trends is still continuing.

Built upon the experience and knowledge gained from this study, a second multinational project, called Diabetes Mondiale (DIAMOND), was launched in January 1990. This will monitor the incidence of and mortality from diabetes in childhood up to the year 2000. Regional training courses in epidemiology and public health aspects of diabetes, and a study of the economic implications of childhood diabetes will also be undertaken as part of the DIAMOND project.

Between 1964 and 1985, WHO convened two expert committees and one study group on diabetes mellitus. These provided a clear account of progress in diabetes research, edu-

cation, and health care over the last two decades. Their findings and recommendations were published in WHO's Technical Report Series. They studied the natural history of the disease, its prevention and control, as well as conditions of employment of diabetics, and diabetes research; and they introduced guidelines for the diagnosis and classification of the disease, including the inadequately understood form of malnutrition-related diabetes mellitus.

Wanted: diabetes education

Considering the importance of public education in any health activity and especially in primary health care, there have been to date comparatively few such programmes in the field of diabetes, and little is yet known about their potential impact. There is certainly scope for demonstration projects aimed at testing the efficacy of public education campaigns in severely affected communities. Furthermore, several components of the integrated approach for the prevention of non-communicable diseases are also of relevance to diabetes.

Educational activities may be aimed not only at the public, but also at non-specialized and specialized health workers. WHO has provided active support and sponsorship for a number of paramedical training courses on diabetes, many of them organized and coordinated by WHO collaborating centres.

In 1981, 1983, 1987 and 1990, international seminars for health professionals on the epidemiology and public health aspects of diabetes have been organized by WHO and co-sponsored by the IDF. A subregional seminar was held in Africa in 1989.

Children learning how to check their own blood glucose levels in a Finnish diabetes centre. This training will make it easier for them to handle their condition throughout their lives.

In progress: national diabetes programme development

Along with the development of guidelines for national diabetes programme development, WHO is also trying to involve professionals of sectors outside the health field with a view to training national teams responsible for the diabetes programmes in their countries. These teams will consist of the national programme coordinator or manager, epidemiologists and biostatisticians, health educators and other key health staff. The first of these intersectoral courses is planned to take place in Geneva this year.

Looking ahead

Diabetes is a growing threat to global public health, to which WHO is paying increasing attention. Our strategy is based mainly on the strengthening of national health services, research and education. It will be essential to adapt the guidelines for diabetes programme development to the local situation, to build on the framework of existing health care systems, and develop appropriate technology. While WHO can act as a catalyst for these activities, national health services should aim at self-reliance in the long term. ■

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